



Artificial intelligence applications in OCT and OCTA for diabetic retinopathy: A systematic review

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ABSTRACT

Purpose: Diabetic retinopathy (DR) is a leading cause of vision impairment worldwide. Optical coherence tomography (OCT) and OCT angiography (OCTA) provide detailed retinal imaging, enabling early detection of microvascular changes. This study aims to systematically review artificial intelligence (AI), particularly deep learning (DL), applications for DR detection and analysis using OCT and OCTA images.

Methods: A comprehensive literature search was conducted across PubMed, Web of Science, Scopus, IEEE Xplore, and Embase for studies published up to March 2026. A total of 1007 articles were identified, of which 175 studies met the inclusion criteria following the PRISMA study selection process.

Results: DL-based approaches consistently demonstrated superior performance compared to traditional machine learning (ML) methods, with reported AUC values typically ranging from 0.90 to 0.99 across classification and segmentation tasks. Convolutional neural networks (CNNs), Vision Transformers (ViTs), and encoder-decoder architectures such as U-Net showed strong performance in detecting key DR biomarkers, including microaneurysms, macular edema, and neovascularization. However, performance variability was observed depending on dataset size, imaging modality, and annotation quality.

Conclusions: AI-driven analysis of OCT and OCTA images offers significant potential for automated DR detection. Despite promising results, challenges such as limited public datasets, lack of cross-institutional validation, and model interpretability remain. Future research should focus on multimodal integration, explainable AI, and large-scale validation to enhance clinical applicability.

1. Introduction

Diabetic retinopathy (DR) is a common microvascular complication of diabetes that affects the retinal vasculature and represents a leading cause of vision impairment and blindness worldwide (Flaxman et al., 2017). The global number of individuals with diabetes is projected to increase from 415 million in 2015 to 642 million by 2040 (Ogurtsova et al., 2017), and the prevalence of DR among these individuals has been reported to range from 30% to 45% (Wong et al., 2016). Fig. 1 illustrates the global prevalence of both diabetes and DR with predictions into the future. Given this growing burden, early detection and continuous monitoring of DR using advanced retinal imaging modalities are essential to prevent disease progression and vision loss. Clinically, DR is classified into two major stages: non-proliferative diabetic retinopathy (NPDR) and proliferative diabetic retinopathy (PDR). NPDR represents the early stage and is often asymptomatic, whereas PDR is characterized by pathological neovascularization resulting from retinal ischemia and disease progression. (Sun et al., 2021).

Early and accurate detection of DR is essential for identifying subtle

structural alterations in the retina before the onset of noticeable clinical symptoms. Detecting these early microvascular and morphological changes enables timely therapeutic intervention, which may slow disease progression and reduce the risk of vision loss (ElTanboly et al., 2017). To address this need, several retinal imaging modalities have been developed for screening, diagnosis, and monitoring of DR. Among these, optical coherence tomography (OCT) has gained significant attention due to its ability to provide high-resolution, three-dimensional (3D), depth-resolved visualization of retinal layers, enabling detection of subtle structural abnormalities and microvascular changes (Spaide et al., 2018). OCT operates based on low-coherence interferometry, where depth information is obtained by measuring the time delay and intensity of backscattered light reflected from different retinal structures.

1.1. Clinical signs of diabetic retinopathy in optical coherence tomography images

Before initiating the diagnosis of DR, it is essential to identify the key

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